



**Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.**

UNIT 10 • STANDARD MPP.7

# Math is Playful

Lesson 10-3 • Teacher Copy

**Name:** \_\_\_\_\_

**Date:** \_\_\_\_\_

**Class:** \_\_\_\_\_

## My Notes

Level 2 Standard



### TODAY'S OBJECTIVES

**Content Objective:** I can use patterns and relationships to make sense of the Tower of Hanoi puzzle and think logically about its solutions.

**Language Objective:** I can describe a puzzle strategy using the words pattern, rule, and predict.



See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.



**Fill in each blank as we go. Use the Word Bank to help you.**



### WORD BANK — FILL EACH BLANK WITH THE BEST WORD

puzzle

pattern

pattern rule

predict

organize



Tap any word to see what it means and a picture.

1

— A problem designed to be played with — it has rules, a goal, and room to imagine solutions.

2

— Something that repeats or changes in a predictable way.

3

— A rule that tells you how to get the next value in a pattern from the one before it.

4

— To say what will happen before it happens, using a pattern or reasoning.

5

— To arrange information — often in a table — so patterns become easier to see.

## Write About the Math

Guided ESOL writing

Use the support level you need. Every level answers the same math question.

### 1. Understand the Question

EXPLAIN

**How can a pattern help you imagine the solution?**

**Your job:** Answer the question. Use math evidence. Explain how the evidence proves your answer.

*Responde la pregunta. Usa evidencia matemática. Explica cómo la evidencia demuestra tu respuesta.*

## 2. Plan Your Math Words

Check at least two words you will use.

- ☐ **puzzle** — A problem designed to be played with — it has rules, a goal, and room to imagine solutions.  
*rompecabezas* — Un problema diseñado para jugarse: tiene reglas, una meta y espacio para imaginar soluciones.

- ☐ **pattern** — Something that repeats or changes in a predictable way.  
*patrón* — Algo que se repite o cambia de una manera predecible.

- ☐ **pattern rule** — A rule that tells you how to get the next value in a pattern from the one before it.  
*regla del patrón* — Una regla que te dice cómo obtener el siguiente valor de un patrón a partir del anterior.

- ☐ **predict** — To say what will happen before it happens, using a pattern or reasoning.  
*predecir* — Decir lo que va a pasar antes de que pase, usando un patrón o el razonamiento.

- ☐ **organize** — To arrange information — often in a table — so patterns become easier to see.  
*organizar* — Ordenar la información, muchas veces en una tabla, para que los patrones sean más fáciles de ver.

**Say it first:** Say your idea to a partner or quietly to yourself before you write.

*Di tu idea a un compañero o en voz baja antes de escribir.*

**The pattern rule for the Tower of Hanoi is to \_\_\_\_ the number of steps for the smaller tower and \_\_\_\_ for each new disc. Applying this rule twice starting from 7 steps for a three-disc tower gives a five-disc tower a minimum of \_\_\_\_ steps.**

*Di tu respuesta primero: Mi respuesta es \_\_\_\_ porque \_\_\_\_.*

### 3. Build Your Explanation

**Model the parts:** This model shows the parts of an explanation. It does not answer your writing question.

**Claim:** The pattern rule is to double the previous number of moves and add 1:  $1 \times 2 + 1 = 3$ , and  $3 \times 2 + 1 = 7$ .

**Evidence:** A strong answer states 'double and add 1' and checks it against both jumps (1→3 and 3→7). A weak answer only notices the numbers are growing without naming the rule.

**Reasoning:** This evidence connects the definition of puzzle to the mathematical claim.

#### Start

Most language support

Complete the frames. Use the word bank.

*Completa las oraciones. Usa el banco de palabras.*

- The pattern rule for the Tower of Hanoi is to \_\_\_\_ the number of steps for the smaller tower and \_\_\_\_ for each new disc. Applying this rule twice starting from 7 steps for a three-disc tower gives a five-disc tower a minimum of \_\_\_\_ steps.

- My answer is \_\_\_\_.

*Mi respuesta es \_\_\_\_.*

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#### Build

Some language support

Write two connected sentences. Use because, so, or but.

*Escribe dos oraciones conectadas. Usa porque, entonces o pero.*

- My claim is \_\_\_\_.

*Mi afirmación es \_\_\_\_.*

- I know because \_\_\_\_.

*Lo sé porque \_\_\_\_.*

- So \_\_\_\_.

*Entonces \_\_\_\_.*

## Explain

### Light language support

Write a claim, give math evidence, and explain your reasoning.

*Escribe una afirmación, da evidencia matemática y explica tu razonamiento.*

- **My claim is \_\_\_\_.**  
*Mi afirmación es \_\_\_\_.*
- **My math evidence is \_\_\_\_.**  
*Mi evidencia matemática es \_\_\_\_.*
- **This proves \_\_\_\_ because \_\_\_\_.**  
*Esto demuestra \_\_\_\_ porque \_\_\_\_.*

## 4. Check Your Explanation

- ☐ I answered the question.  
*Respondí la pregunta.*
- ☐ I used math evidence: numbers, an equation, a model, or a comparison.  
*Usé evidencia matemática: números, una ecuación, un modelo o una comparación.*
- ☐ I explained how the evidence proves my answer.  
*Explicué cómo la evidencia demuestra mi respuesta.*
- ☐ I used at least two math words.  
*Usé por lo menos dos palabras matemáticas.*
- ☐ I reread complete sentences and punctuation.  
*Volví a leer mis oraciones completas y la puntuación.*

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## Writing Guide — Teacher Copy

The support level changes the amount of language scaffolding, not the mathematical expectation. Look for the same five features in every response.

- I answered the question.
- I used math evidence: numbers, an equation, a model, or a comparison.
- I explained how the evidence proves my answer.
- I used at least two math words.
- I reread complete sentences and punctuation.

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## Answer Key & Teacher Guide

1. **Notes 1:** puzzle
2. **Notes 2:** pattern
3. **Notes 3:** pattern rule
4. **Notes 4:** predict
5. **Notes 5:** organize
6. **Watch for this mistake:** *Doubling the previous number of steps but forgetting to add 1 — answering 14 instead of 15 for four discs, or 30 instead of 31 for five.*
7. **Try It 1:**
8. **Try It 2:**